ST JOSEPH ENGINEERING COLLEGE MANGALURU

AN AUTONOMOUS INSTITUTION

The Department of Intelligent Computing and Business Systems

Course Level Problem Based Learning

Course Title: Database Management System

Course Code: 22AIM43

Semester: Fourth Semester

A CLPBL Project Report Titled

Book Bridge

Submitted By

Sl No.	USN	Name
1	4SO24AI007	Anson Rodrigues
2	4SO24AI026	Kavya D Marla
3	4SO24AI028	Mahammad Arfaz
4	4SO24AI030	Manish G Suvarna
5	4SO24AI042	Navita Vinod Shetty
6	4SO24AI059	Shaun Jayson D’Souza

Under the Guidance of

Dr Saleena T S
Assistant Professor

13 June 2026

ST JOSEPH ENGINEERING COLLEGE MANGALURU – 575 028

The Department of Intelligent Computing and Business Systems
CERTIFICATE OF COMPLETION

It is hereby certified that this CLPBL Project Report is a bonafide work carried out by the students listed below under my guidance in the Department of Intelligent Computing and Business Systems. This work has been completed towards the partial fulfilment of the course Database Management System with course code 22AIM43 offered in the Third Semester.

It is further certified that the students have followed the Design Thinking framework throughout the execution of the project and have incorporated all the corrections/suggestions provided during the progress reviews and assessments. The project work has been evaluated and approved as it meets the requirements of CLPBL Implementation as mandated by the Department.

Sl No.	USN	Name
1	4SO24AI007	Anson Rodrigues
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5	4SO24AI042	Navita Vinod Shetty
6	4SO24AI059	Shaun Jayson D’Souza

Dr Saleena T S
Assistant Professor
Department of ICBS

Dr Hari Vinod
Head of the Department
Department of ICBS

ST JOSEPH ENGINEERING COLLEGE MANGALURU – 575 028

The Department of Intelligent Computing and Business Systems

DECLARATION

We, the undersigned students of the Department of Intelligent Computing and Business Systems, hereby declare that the project work entitled **"BookBridge"** submitted in partial fulfilment of the requirements of the course **Database Management System** with course code **22AIM43**, in the **Fourth Semester**, is an original **Course Level Problem Based Learning (CLPBL)** project carried out by us.

We confirm that this project has been completed using the **Design Thinking framework** under the guidance of **Dr Saleena T S**. We also certify that the work reported in this document has not been previously submitted to any institution or organization for any academic award, degree, or certification. We abide by the academic integrity and ethical standards of the Department and the Institution, and accept that any violation, if discovered later, may lead to appropriate action.

Students' Signatures:

Sl No.	USN	Name	Signature
1	4SO24AI007	Anson Rodrigues	
2	4SO24AI026	Kavya D Marla	
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5	4SO24AI042	Navita Vinod Shetty	
6	4SO24AI059	Shaun Jayson D'Souza	

Date: 13 June 2025

Place: Mangaluru

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We are immensely thankful to the Head of the Department, Dr Hari Vinod, for providing the necessary support and a conducive learning environment.

We also extend our heartfelt thanks to the CLPBL Implementation Committee and all the faculty members who reviewed our project periodically and shared their suggestions for improvement.

We acknowledge the institutional support provided by St Joseph Engineering College – Mangaluru for enabling us to apply the Design Thinking framework and explore real-world solutions through this project.

Finally, we thank all our classmates, respondents, users, and stakeholders who extended their cooperation and contributed to the successful completion of our project.

Team Members:

Sl No.	USN	Name
1	4SO24AI007	Anson Rodrigues
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5	4SO24AI042	Navita Vinod Shetty
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ABSTRACT

This Course Level Problem Based Learning (CLPBL) project focused on addressing the challenge of high textbook costs and the inefficient exchange of secondhand academic books among students. Many students purchase textbooks for a single semester and are left with unused books, while others struggle to find affordable study materials. Existing secondhand marketplaces are often unstructured, lack transparency, and do not provide reliable mechanisms for condition verification, fair pricing, or negotiation tracking. Motivated by the need to create a cost-effective and student-friendly solution, the team adopted the Design Thinking framework to guide the development process. Through empathy mapping and problem analysis, key challenges were identified, including difficulties in locating relevant textbooks, assessing book quality, communicating with sellers, and managing negotiations. Ideation sessions led to the development of a dedicated marketplace prototype named Book Bridge. The platform was designed with smart listings that include detailed book information, condition grading, and images, along with a structured offer and counteroffer system that maintains complete negotiation history. Additionally, the platform integrates secure third-party payment gateways to facilitate safe and transparent transactions. User feedback highlighted the platform's ease of use, improved trust between buyers and sellers, and its effectiveness in reducing textbook expenses. The project outcomes demonstrated how a student-centric digital marketplace can streamline textbook exchange, promote affordability, and enhance transparency within educational communities. Overall, Book Bridge showcased the potential of technology-driven solutions to support sustainable resource sharing and improve accessibility to academic materials.

Keywords: Problem Based Learning, Design Thinking, Secondhand Marketplace, Textbook Exchange, Student Community, Structured Negotiation, Digital Platform.

Table of Contents

Chapter – 1: Problem Description

The increasing cost of academic textbooks has become a major challenge for students pursuing higher education. Many students are required to purchase expensive textbooks for a single semester, after which these books often remain unused. At the same time, other students seek affordable alternatives but face difficulties in finding relevant secondhand books at reasonable prices. This problem primarily affects college and university students who depend on textbooks for their academic studies and are looking for cost-effective learning resources. The issue also indirectly affects educational institutions by limiting access to affordable study materials and reducing opportunities for resource reuse. The challenge is particularly evident in college environments where students frequently buy and sell textbooks through informal channels such as social media groups, messaging platforms, or local marketplaces. These methods often lack organization, transparency, and reliability. Buyers face difficulties in verifying the condition, edition, and authenticity of books, while sellers struggle to reach the appropriate audience and receive fair value for their books. The absence of structured negotiation mechanisms further complicates transactions, leading to confusion, disputes, and inefficient communication. As a result, students may spend more than necessary on textbooks or fail to obtain required study materials on time. The problem extends beyond financial concerns and contributes to the underutilization of educational resources, as many usable textbooks remain unused instead of being circulated among students. This challenge is commonly observed in colleges and universities where there is no dedicated platform for managing secondhand textbook exchanges. With the rising cost of education and increasing demand for affordable learning resources, the need for a transparent, organized, and student-focused marketplace has become increasingly important. Addressing this issue can improve textbook accessibility, reduce student expenses, promote sustainable resource sharing, and create a more efficient ecosystem for academic book exchange.

Chapter – 2: Key Findings of Secondary Research

A review of secondary literature was conducted using research papers, academic repositories, and case studies related to secondhand marketplaces and student-focused trading platforms. The study examined existing systems for buying and selling pre-owned goods, user trust in online marketplaces, and factors influencing successful peer-to-peer transactions.

A research paper titled *"A Sustainable Platform for Buying and Selling Pre-owned Goods Among College Students: Campus Bazaar"* published in the International Journal for Research in Applied Science & Engineering Technology (IJRASET) highlighted the benefits of dedicated student marketplaces in promoting affordability, sustainability, and resource reuse. The study emphasized that college students often face financial challenges when purchasing academic resources and can benefit significantly from organized secondhand exchange platforms.

Research available through the MIT DSpace repository explored user behavior and trust mechanisms in online marketplaces. The findings indicated that transparency, accurate product information, and reliable communication are critical factors influencing user participation and transaction success. Buyers were more willing to engage in transactions when sufficient information about product condition and seller credibility was available.

Further studies from California State University and Queensland University of Technology examined peer-to-peer trading platforms and online marketplace usability. These studies highlighted common challenges such as unclear product descriptions, inefficient communication, lack of negotiation tracking, and trust issues between buyers and sellers. The research also found that structured interactions, seller ratings, and transaction histories improve user confidence and platform reliability.

The literature review identified several opportunities for improvement in textbook exchange systems. These include implementing detailed book condition grading, structured offer and counteroffer mechanisms, efficient search and filtering options, secure payment integration, and transaction tracking features. Mobile-friendly design and student-focused functionality were also identified as important factors for increasing adoption among college students.

Overall, the secondary research confirmed the need for a dedicated textbook marketplace that addresses affordability, transparency, trust, and transaction efficiency. These findings provided the foundation for developing Book Bridge, a student-centric platform designed to simplify the buying and selling of secondhand academic textbooks.

Chapter 3: Key Findings of Primary Research

Primary research for the CLPBL project was conducted using a Google Form survey circulated among undergraduate students, competitive exam aspirants, and individuals involved in buying and selling secondhand textbooks. The objective was to understand the challenges students face in textbook exchange, pricing, trust, and accessibility of affordable academic resources.

Survey Observations and Stakeholder Feedback

The survey revealed that students mainly rely on friends, seniors, WhatsApp groups, and online marketplaces to buy and sell used textbooks. However, these methods often lack transparency and reliability. About **75.2%** of respondents reported difficulty finding the required books, while **45.2%** had been misled about book condition. Additionally, **32.8%** experienced failed transactions due to trust issues or unclear agreements. The average trust rating for informal sellers was only **2.77 out of 5**.

Respondents also highlighted challenges such as difficulty finding the correct edition, lack of seller verification, and the absence of a proper platform to compare prices and manage transactions.

User Needs, Pain Points, and Expectations

The primary research identified the following key needs:

• A dedicated platform for buying and selling academic textbooks.

• Accurate information about book condition, edition, and pricing.

• Verified student accounts to improve trust and security.

• Structured negotiation and transaction tracking.

• Efficient search and filtering options.

• Dispute-resolution mechanisms and transparent communication.

The findings further showed that **72.8%** of respondents would use a dedicated textbook marketplace if available, confirming the need for **Book Bridge**, a student-centric platform that promotes affordability, transparency, and efficient textbook exchange.

Empathy Map Insights The Empathy Maps prepared during primary research further reinforced these findings.

Section	Guiding Questions	Student Notes / Observations

1. Who are we empathizing with?	Who is the user? What is their role or situation? What decisions do they make?	Primary Users: Students who want to sell used textbooks and buy affordable secondhand textbooks. Secondary Users: Seniors selling old semester books and Juniors searching for required textbooks. They decide whether to buy, sell, accept, reject or negotiate the textbook price.
2. What do they need to do?	What tasks or goals are they trying to accomplish? What problems are they trying to solve?	To buy or sell textbook quickly at a fair and affordable price. High textbook costs, difficulty finding trusted buyers or selling and unclear book conditions.
3. What do they see?	What environment are they in? What do they see in the market or surroundings?	A college or online student marketplace where textbooks are exchanged between students. Expensive new textbooks, many secondhand listings, different book conditions and varying prices from sellers.
4. What do they say?	What have you heard them say about the problem? What words or stories do they use?	“New textbooks are too expensive and it is hard to find trustworthy sellers with good-quality books.” “I need affordable books quickly but negotiating prices and checking book condition takes too much time.”

5. What do they do?	What actions or behaviours have you observed?	Comparing prices, checking book condition, negotiating with sellers, and browsing multiple listings before making a decision.
6. What do they hear?	What do friends, colleagues, or influencers say? What sources do they listen to?	Friends, classmates and seniors often suggest trusted sellers, recommend affordable books and share their experiences about successful or bad transactions. Students also rely on social media groups, online marketplace reviews and peer recommendations before deciding to buy or sell textbooks.
7. What do they think and feel?	What keeps them awake at night? What are their fears or dreams?	They worry about paying too much, getting books in poor condition, being scammed or not finding the required textbook before classes or exams. They want an easy, trustworthy and affordable way to buy or sell textbooks quickly without stress.
8. Pains	What frustrations, obstacles, or risks do they face?	Users face unclear book conditions, fake or unresponsive buyers and sellers, long negotiating processes, price disagreements, and the risk of scams or delayed transactions.

9. Gains	What does success look like? What benefits do they desire?	Successfully buying or selling textbooks quickly at a fair price with a smooth and trustworthy transaction. Affordable prices, reliable users, clear book condition details, easy negotiation, and time-saving transactions.

Chapter - 4: Problem Definition

The increasing cost of academic textbooks has become a significant challenge for students pursuing higher education. Many students are required to purchase multiple textbooks each semester, resulting in substantial educational expenses. While these books are often used for only a short duration, they frequently remain unused after the completion of a course. At the same time, other students struggle to access affordable study materials and are forced to either purchase expensive new books or rely on limited library resources. This situation creates both financial and accessibility challenges within academic communities.

Existing methods for buying and selling secondhand textbooks are often unstructured and inefficient. Students commonly depend on informal channels such as friends, seniors, social media groups, messaging platforms, and general online marketplaces. Although these platforms facilitate textbook exchange to some extent, they are not specifically designed to meet the needs of students. Buyers often face difficulties in locating the correct title, edition, or condition of a book, while sellers struggle to reach interested buyers within their academic community. As a result, many potentially useful textbooks remain unused despite existing demand.

A major concern identified through primary and secondary research is the lack of transparency in book listings. Information regarding book condition, edition, and pricing is often incomplete or inaccurate, making it difficult for buyers to make informed decisions. Many students have reported being misled about the condition of books or receiving materials that do not meet their expectations. This lack of reliable information reduces trust between buyers and sellers and discourages participation in secondhand textbook exchanges.

Another significant challenge is the absence of a structured negotiation process. Price discussions typically occur through private messages, phone calls, or external communication platforms. These informal negotiations provide no mechanism for tracking offers, counteroffers, or agreed prices. Consequently, misunderstandings, delayed responses, and failed transactions are common. Students frequently encounter difficulties in reaching fair agreements, resulting in frustration and inefficiencies during the buying and selling process.

The issue is further complicated by the lack of trust and accountability in existing systems. Most informal marketplaces do not provide seller verification, transaction history, or reliable dispute-resolution mechanisms. Buyers often hesitate to purchase books from unknown sellers, while sellers are concerned about fraudulent inquiries and unreliable buyers. The absence of trust-building features creates barriers to successful transactions and limits the effectiveness of textbook exchange platforms.

This challenge is particularly relevant during semester transitions when demand for academic books is highest. Students require quick access to affordable textbooks to support their studies, yet the existing exchange methods often fail to provide a seamless experience. Delays in communication, difficulty in finding required books, and uncertainty regarding product quality contribute to a poor user experience and increase educational costs.

The implications of this problem extend beyond individual students. High textbook costs can reduce access to educational resources, particularly for students with financial constraints. In addition, the underutilization of secondhand books contributes to unnecessary waste and limits opportunities for sustainable resource sharing within educational institutions. An effective textbook exchange system can therefore improve affordability, accessibility, and environmental sustainability simultaneously.

In summary, students face significant challenges in buying and selling secondhand textbooks due to unorganized marketplaces, lack of transparency, trust issues, inefficient negotiation methods, and inadequate transaction management. Existing solutions address only part of the problem and fail to provide a dedicated student-focused environment. There is a clear need for a specialized platform that enables detailed book listings, structured negotiation, secure transaction tracking, and trusted interactions. Addressing these challenges can reduce educational expenses, improve access to academic resources, and create a more efficient and transparent textbook exchange ecosystem for students.

Chapter - 5: Project Objectives

The objectives of this CLPBL project were carefully formulated to address the challenges faced by students in buying and selling secondhand academic textbooks. The rising cost of textbooks often creates a financial burden on students, while the lack of a dedicated and reliable marketplace makes textbook exchange difficult and inefficient. Based on insights gathered through primary research, secondary research, empathy mapping, and stakeholder feedback, the project aims to improve affordability, accessibility, transparency, trust, and overall transaction efficiency. The objectives are aligned with the Design Thinking framework and focus on creating a practical, user-centered solution that addresses real student needs.

The primary objective of the project was to design and develop **Book Bridge**, a dedicated student-centric marketplace that enables students to buy and sell secondhand textbooks in a simple, convenient, and organized manner. Existing methods such as WhatsApp groups, social media platforms, and informal student networks often lack structure and reliability. Therefore, the project seeks to provide a centralized platform where students can easily connect with buyers and sellers within their academic community.

Another important objective was to develop a **smart listing system** that allows sellers to provide complete and accurate information about textbooks. The platform enables users to include details such as title, author, edition, subject, condition, price, and images. This objective addresses the common problem of incomplete or misleading information and helps buyers make informed purchasing decisions.

A key objective of the project was to implement a **structured negotiation mechanism** that supports offers and counteroffers between buyers and sellers. Research findings revealed that price negotiations are usually conducted through private messages, leading to confusion and lack of transparency. By maintaining a negotiation history, the platform aims to promote fair pricing, improve communication, and create a more organized transaction process.

The project also aims to enhance **trust and credibility** within the marketplace. Many students expressed concerns regarding unverified users, inaccurate book descriptions, and failed transactions. To address these issues, the platform focuses on transparency, detailed listings, and transaction tracking, creating a trustworthy environment for textbook exchange.

Another objective was to integrate **secure transaction tracking** through third-party payment gateways such as Razorpay or Stripe. Although payment processing is handled externally, the platform maintains transaction records and payment status information to improve accountability and reduce disputes between buyers and sellers.

The project further aims to provide **efficient search and filtering capabilities** that allow users to quickly locate textbooks based on subject, title, author, edition, condition, and price range. This objective seeks to reduce the time and effort required to find relevant books and improve overall accessibility to academic resources.

An additional objective was to encourage the **reuse of educational resources** within academic communities. Many textbooks remain unused after a semester despite being valuable to future students. By facilitating the exchange of secondhand books, the platform promotes affordability, sustainability, and better utilization of educational materials.

The project also focuses on conducting **user testing and feedback collection** throughout the development process. Continuous feedback helps identify usability issues, validate design decisions, and ensure that the final solution effectively meets student expectations. This objective supports the development of a platform that is practical, user-friendly, and responsive to stakeholder needs.

Another objective was to demonstrate the effective application of the **Design Thinking framework** through the stages of empathy, problem definition, ideation, prototyping, and testing. By following this approach, the project highlights the importance of user-centered innovation in solving real-world educational challenges.

Finally, the project aims to evaluate the overall impact of the proposed solution by measuring improvements in affordability, accessibility, transparency, trust, and user satisfaction. It also explores future enhancements such as student verification systems, seller ratings, dispute-resolution mechanisms, and expansion to multiple educational institutions.

In summary, the objectives of Book Bridge focus on creating a transparent, affordable, and reliable textbook marketplace that simplifies textbook exchange, promotes trust, encourages resource reuse, and improves access to academic materials. Through a combination of user-centered design, structured negotiation, secure transaction tracking, and continuous improvement, the project aims to provide meaningful value to students and educational communities.

Chapter - 6: Business Model Canvas

Business Model Canvas

Designed for:	Designed by:	Date:	Version:
Students	Phoenix	13/06/2006	

Key Partners

- SJEC Administration
- Students (buyers & sellers)
- Google OAuth
- College departments
- Hosting providers
- Payment/meeting coordination partners

Key Activities

- Textbook listing management
- User verification
- Negotiation handling
- Messaging & notifications
- Dispute resolution
- Platform maintenance

Key Resources

- Web platform
- Student user base
- MySQL database
- Authentication system
- Book listings data
- Development team

Value Propositions

- Affordable second-hand textbooks
- Verified SJEC-only marketplace
- Secure seller verification
- Multi-buyer negotiation system
- Campus-focused trust & safety
- Easy textbook discovery

Customer Relationships

- Self-service platform
- In-app messaging
- Notifications & updates
- Reviews and ratings
- Dispute support

Channels

- Web application
- Google OAuth login
- Campus word-of-mouth
- Student communities
- Department groups

Customer Segments

- SJEC students
- Book buyers
- Book sellers
- Freshers seeking textbooks
- Seniors selling used books

Cost Structure

- Hosting & infrastructure
- Domain and deployment costs
- Database maintenance
- Development effort
- Platform monitoring
- Support & maintenance

Revenue Streams

- Featured listings (future)
- Premium seller tools (future)
- Campus partnerships
- Advertising (future)
- Commission-free MVP model

Chapter - 7. SWOT Analysis

The SWOT analysis was conducted to evaluate the strengths, weaknesses, opportunities, and threats associated with the proposed solution, Book Bridge.

Strengths

1. Book Bridge is a dedicated student-centric platform specifically designed for secondhand textbook exchange.
2. Smart listings provide detailed information such as title, edition, condition, price, and images, improving transparency.
3. The structured negotiation system enables buyers and sellers to exchange offers and counteroffers while maintaining negotiation history.
4. Transaction tracking enhances accountability by maintaining records of exchanges and payment status.
5. The platform promotes affordability and sustainability by encouraging the reuse of academic resources.

Weaknesses

1. The effectiveness of the platform depends on active student participation and the availability of textbook listings.
2. Information accuracy relies on sellers providing honest and complete details about their books.
3. Limited user adoption may reduce the variety and availability of textbooks.
4. Dependence on third-party payment gateways may affect transaction processing.
5. Continuous maintenance, technical support, and updates are required to ensure smooth platform operation.

Opportunities

1. The growing demand for affordable educational resources creates a strong market opportunity.
2. The platform can be expanded to multiple colleges and universities, increasing its reach and impact.

3. Features such as student verification, seller ratings, and reviews can further enhance trust and credibility.
4. AI-based recommendations and fair price suggestions can improve user experience.
5. Partnerships with educational institutions and student organizations can support wider adoption and growth.

Threats

1. Competition from existing online marketplaces and social media-based textbook exchange groups.
2. Fraudulent listings, misleading descriptions, or disputes between buyers and sellers may reduce user trust.
3. Data privacy and security concerns related to user information and transaction records.
4. Resistance from students who are already comfortable using existing exchange methods.
5. Technical issues such as server downtime, software bugs, or payment gateway failures may affect platform reliability.

Overall, the SWOT analysis indicates that Book Bridge has strong potential to improve textbook accessibility and affordability while creating a transparent and efficient marketplace for students.

HELPFUL	HARMFUL

Strength:	**Weakness:**
1.Student-Centric Design	1. Dependence on User Participation
2.Trusted and Verified Community	2. Dependence on Internet Connectivity
3.Structured Negotiation System	3. Limited Scope Initially
4.Trusted and Verified Community	4. Limited Initial Inventory
Opportunity:	**Threats:**
1.Expansion to Multiple Colleges	1.Competition from Existing Platforms
2.Integration with College Ecosystems	2.Changes in Learning Methods
3.Competitive Exam Book Marketplace	3.Market Saturation
4.Digital Learning Resource Exchange	4.Dependence on Academic Cycles

Chapter – 8: Description of Work

Empathize Phase: Understanding the User

The Empathize phase focused on understanding the challenges faced by students when buying and selling secondhand academic textbooks. During this phase, the team conducted secondary research using research papers, online articles, academic reports, and studies related to student marketplaces and peer-to-peer trading platforms. The objective was to understand existing textbook exchange practices, common user difficulties, and factors influencing trust and transaction success.

In addition, primary research was conducted through Google Form surveys and discussions with students from different academic disciplines. The responses revealed that students often struggle to find required textbooks at affordable prices and frequently rely on informal channels such as friends, seniors, WhatsApp groups, and social media platforms. Many respondents reported concerns regarding inaccurate book descriptions, difficulty in finding the correct edition, lack of trust in sellers, and inefficient negotiation processes.

To consolidate these findings, the team created Empathy Maps that captured what users say, think, feel, and do during textbook transactions. The analysis highlighted key pain points such as high textbook costs, lack of transparency, trust issues, poor discoverability of books, and unstructured communication. The Empathize phase helped the team gain a deeper understanding of user needs and establish a strong foundation for problem definition.

Define Phase

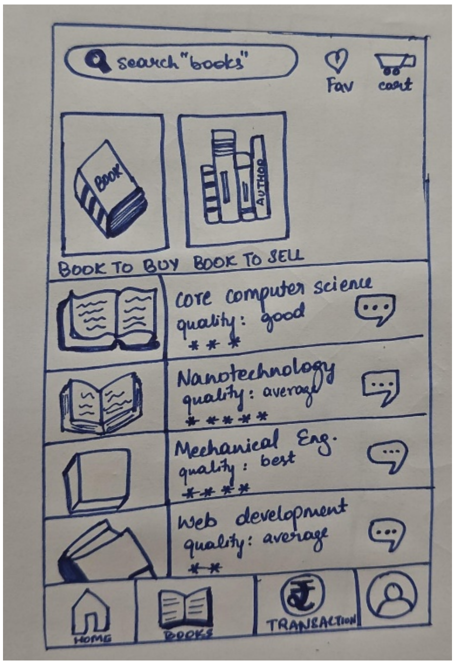
Based on insights gathered during the Empathize phase, the team clearly defined the core problem. Existing methods for buying and selling secondhand textbooks were found to be fragmented, unreliable, and lacking features specifically designed for students.

The problem was identified as the absence of a dedicated platform that enables students to easily discover textbooks, verify book condition, negotiate prices transparently, and complete transactions efficiently. Existing solutions failed to provide structured negotiation, trust-building mechanisms, and transaction tracking. This phase helped the team develop a focused problem statement and ensured that all subsequent design decisions were aligned with addressing student needs.

8.1 Ideation and Design

The ideation and design phase was conducted using the Design Thinking framework with a strong emphasis on creating a student-friendly and transparent marketplace. Based on research findings, several brainstorming sessions were conducted to explore potential solutions for textbook exchange.

Various approaches were evaluated, including social media-based groups, classified advertisement systems, chat-based marketplaces, and dedicated textbook exchange platforms. Each alternative was assessed based on usability, affordability, scalability, transparency, and ease of adoption by students.



Following evaluation, the team selected a dedicated web-based marketplace solution named **Book Bridge**. The platform was designed around three primary features: Smart Listings, Structured Negotiation, and Transaction Tracking.

Key design decisions included:

- Detailed book listings containing title, author, edition, condition, price, and images.
- Search and filtering functionality for quick book discovery.
- Offer and counteroffer mechanisms with negotiation history.

- Transaction tracking and payment status monitoring.
- Simple and intuitive user interface suitable for students.
- Mobile-responsive design for accessibility across devices.

Low-fidelity wireframes and interface sketches were developed and refined through multiple iterations. Feedback from students was incorporated to improve navigation, listing presentation, and negotiation workflows. This phase resulted in a clearly defined system architecture and interface design that guided prototype development.

8.2 Prototyping / Simulation

The proposed solution, **Book Bridge**, was developed as a functional web-based prototype that simulates real-world textbook buying and selling scenarios. The prototype was designed to demonstrate the platform's core functionality and validate user workflows.

Prototype Description

The prototype consists of two primary user modules:

Seller Module

- Create and manage textbook listings.
- Upload book details and images.
- Receive and respond to offers.
- Track negotiation history and transaction status.

Buyer Module

- Search and browse available textbooks.
- Apply filters based on subject, edition, price, and condition.
- Submit offers and negotiate with sellers.
- View transaction records and purchase history.

The system simulates the complete textbook exchange process from listing creation to negotiation and transaction completion.

Software Tools and Technologies Used

The frontend was developed using HTML, CSS, JavaScript, and React.js to create a responsive and interactive user interface. The backend was implemented using Node.js and Express.js for handling business logic and API communication. Data storage was managed using MongoDB/Firebase databases. Payment integration was simulated through third-party payment gateways such as Razorpay or Stripe. The platform was designed to support cloud-based deployment and future scalability.

Implementation Approach

The prototype follows a modular architecture consisting of presentation, application, and database layers. User data, book listings, negotiation records, and transaction details are stored separately to ensure efficient data management. APIs facilitate communication between frontend and backend components, enabling smooth user interaction and real-time updates.

8.3 Testing

Testing was conducted to evaluate the functionality, usability, and reliability of the Book Bridge prototype. The objective was to verify that the system performed according to the requirements identified during research and design phases.

Testing Approach

The testing process included:

- Functional Testing
- Usability Testing
- Scenario-Based Testing
- Interface Testing

- Transaction Workflow Testing

Parameters Evaluated

The following parameters were evaluated during testing:

- Accuracy of book listing creation and updates.
- Search and filter functionality.
- Offer and counteroffer processing.
- Negotiation history tracking.
- Transaction recording and payment status updates.
- User interface responsiveness across devices.
- Overall ease of navigation and usability.

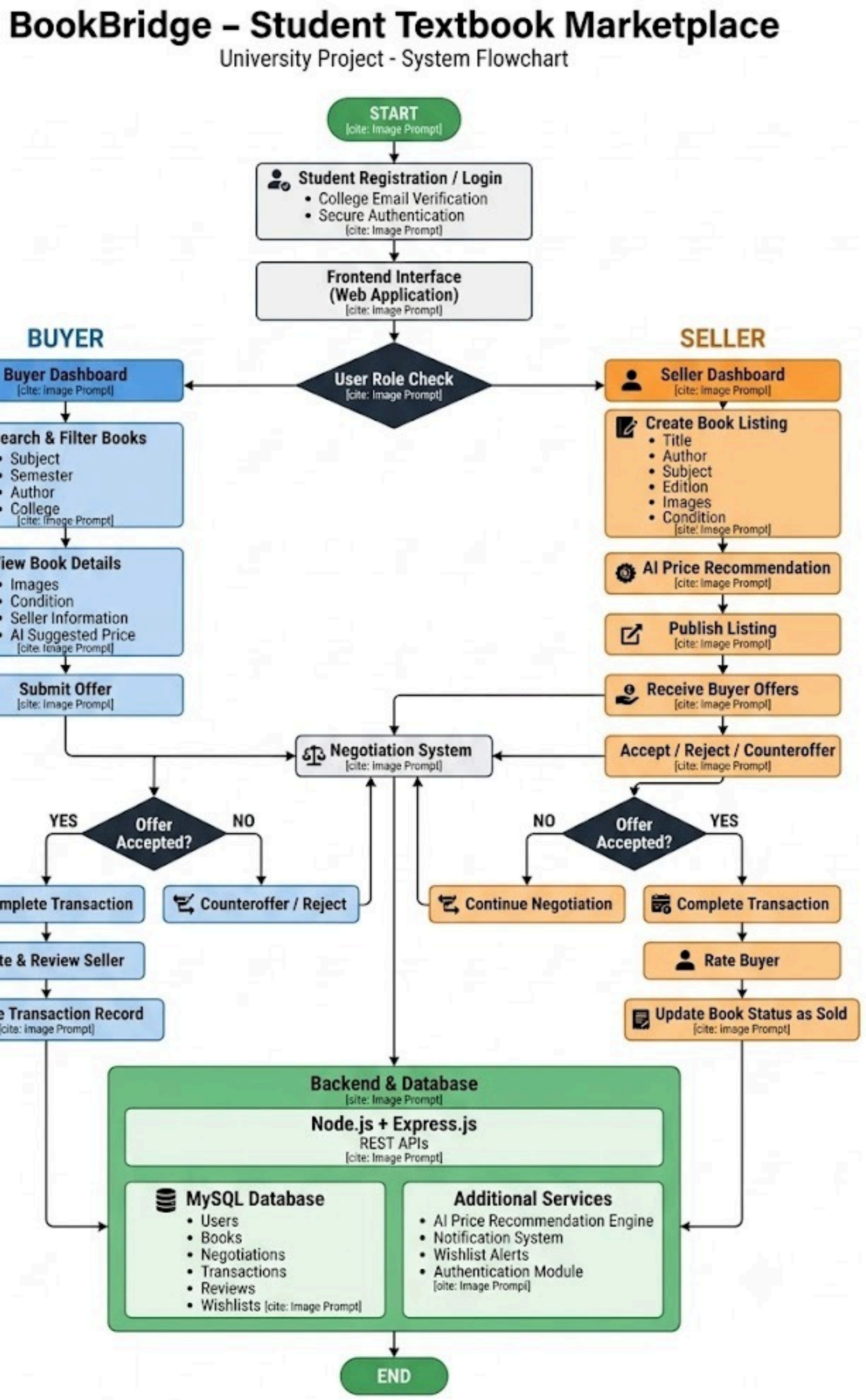
Test Results

The testing results indicated that users were able to create listings, search for books, negotiate prices, and track transactions successfully. Search and filtering features helped users quickly locate relevant textbooks. The negotiation system accurately recorded offers and counteroffers, providing transparency throughout the process. Transaction records were maintained correctly, improving accountability and trust between users.

User Feedback and Improvements

Feedback from students highlighted the usefulness of detailed book information, structured negotiation, and transaction tracking features. Users appreciated the simplicity of the interface and the ability to compare books efficiently. Based on feedback, improvements were made to search filters, listing presentation, and notification mechanisms. Additional recommendations included implementing seller ratings, dispute-resolution systems, and AI-based book recommendations in future versions.

Overall, testing confirmed the feasibility and effectiveness of Book Bridge as a student-centric marketplace capable of improving transparency, affordability, and efficiency in secondhand textbook exchange.


BookBridge – Student Textbook Marketplace
University Project - System Flowchart
START
Student Registration / Login
• College Email Verification
• Secure Authentication
Frontend Interface (Web Application)
BUYER
SELLER
User Role Check
Buyer Dashboard
Search & Filter Books
• Subject
• Semester
• Author
• College
View Book Details
• Images
• Condition
• Seller Information
• AI Suggested Price
Submit Offer
Seller Dashboard
Create Book Listing
• Title
• Author
• Subject
• Edition
• Images
• Condition
AI Price Recommendation
Publish Listing
Receive Buyer Offers
Accept / Reject / Counteroffer
Negotiation System
YES
NO
Offer Accepted?
Complete Transaction
Counteroffer / Reject
Continue Negotiation
Rate & Review Seller
Rate Buyer
Store Transaction Record
Update Book Status as Sold
Backend & Database
Node.js + Express.js
REST APIs
MySQL Database
• Users
• Books
• Negotiations
• Transactions
• Reviews
• Wishlists
Additional Services
• AI Price Recommendation Engine
• Notification System
• Wishlist Alerts
• Authentication Module
END

Chapter – 9: Expenses

This CLPBL project was primarily a software-based solution developed using open-source technologies and free development platforms. As a result, no material procurement or fabrication costs were incurred during the execution of the project. All development, testing, and deployment activities were carried out using freely available tools and resources. The project did not require any physical components, hardware devices, or paid software licenses. Development was performed using personal computers owned by team members, and cloud services were utilized under free-tier plans. Therefore, the overall project cost remained minimal and within academic constraints.

Chapter – 10: Conclusion

The Course Level Problem Based Learning (CLPBL) project on **Book Bridge – Student-Centric Secondhand Textbook Marketplace** provided a valuable opportunity to address a real-world problem faced by students in accessing affordable academic resources. The high cost of textbooks, combined with the lack of an organized and transparent exchange system, often forces students to rely on informal channels that are inefficient and unreliable. By applying the Design Thinking framework, the team was able to empathize with users, define key challenges, ideate practical solutions, and develop a structured prototype that directly addresses these issues.

The final solution, Book Bridge, was designed as a user-friendly platform that enables students to buy and sell secondhand textbooks through smart listings, structured negotiation, and secure transaction tracking. The system ensures transparency by providing detailed book information, improves trust through structured communication between buyers and sellers, and enhances usability through efficient search and filtering features. The project outcomes demonstrated improved accessibility to textbooks, better price transparency, and a more organized exchange process compared to traditional informal methods.

Beyond the immediate solution, the project highlights the importance of digital transformation in educational resource sharing. It demonstrates how technology can be used to reduce educational costs, promote sustainability through reuse of textbooks, and improve collaboration within student communities. The approach also emphasizes the value of user-centered design in building systems that are practical, efficient, and aligned with real user needs.

Future enhancements for Book Bridge include expansion to multiple institutions, integration of student verification systems, seller ratings, AI-based recommendations, and advanced dispute resolution mechanisms. These improvements can further strengthen trust, scalability, and usability of the platform.

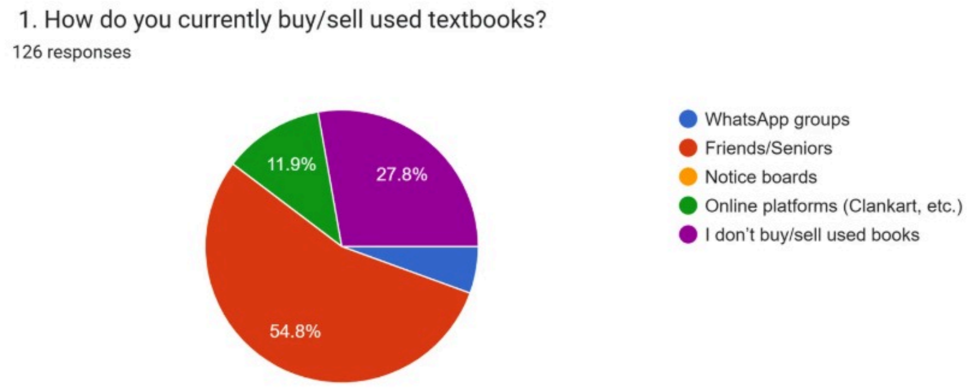
In conclusion, Book Bridge is more than just a prototype—it represents a step toward a smarter, more transparent, and more affordable academic ecosystem. By combining innovation with user-centric design, the project demonstrates how engineering solutions can effectively solve everyday student challenges and contribute to a more efficient and sustainable educational environment.

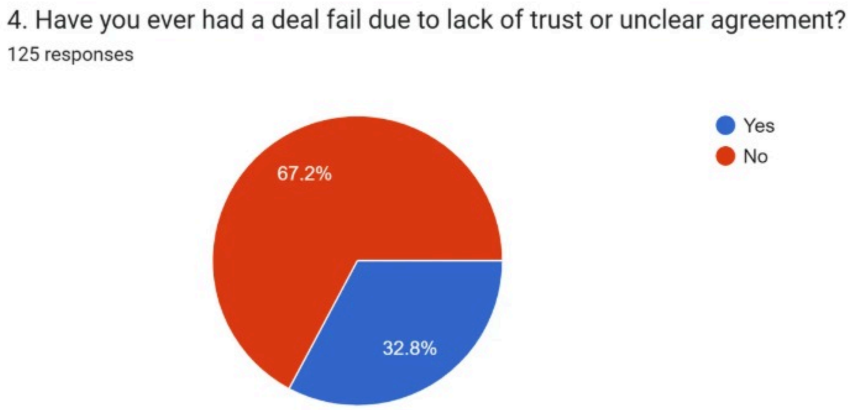
References

dspace.mit.edu/handle/1721.1/66425

https://scholarworks.calstate.edu/downloads/mc87pt14x https://eprints.qut.edu.au/55244/

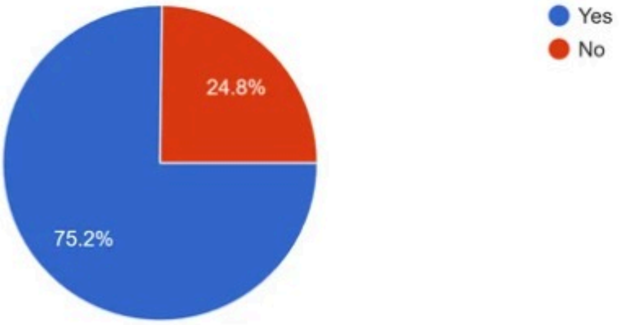
Appendices


1. How do you currently buy/sell used textbooks?
126 responses
11.9%
27.8%
54.8%
WhatsApp groups
Friends/Seniors
Notice boards
Online platforms (Clankart, etc.)
I don't buy/sell used books


4. Have you ever had a deal fail due to lack of trust or unclear agreement?
125 responses
67.2%
32.8%
Yes
No

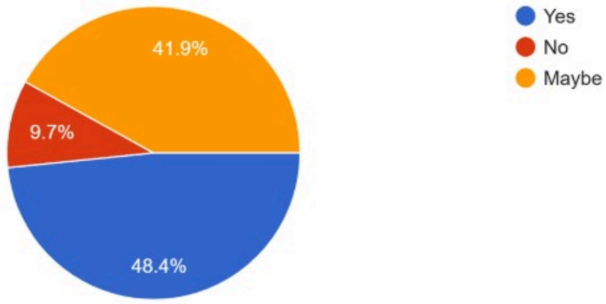
6.Have you ever wanted a book but couldn't find it anywhere?

125 responses



11.Would a system-suggested fair price help you during negotiation?

124 responses



14. Would you use such a platform if it were available for your institution?

125 responses